

GHANA AFLATOXIN DATA ANALYSIS FOR ML MODELLING.

The sampling locations were initially recorded in **Packed Degrees–Minutes–Seconds (Packed DMS)** format. To facilitate compatibility with Geographic Information Systems (GIS), web mapping applications, and spatial analysis software, the coordinates were converted to **decimal degrees (DD)**. The conversion was performed using the standard Degrees–Minutes–Seconds (DMS) to Decimal Degrees conversion equation:

$$\text{Decimal degrees} = \text{Decimal degrees} + (\text{Minutes}/60) + (\text{seconds}/3600)$$

For west longitudes, the resulting decimal degree values were assigned to a negative sign. Following coordinate conversion, all sampling locations were imported into **Google Earth Pro** for quality assurance. The converted coordinates were visually verified against the recorded village and town names to ensure that each sampling location corresponded to its documented geographical location. This verification confirmed the consistency between the converted coordinates and the recorded sampling sites.

N	W/E		Latitude	Longitude
0548294	00020649	W	5.8049	-0.34415
0548294	00020649	W	5.8049	-0.34415
0548294	00020649	W	5.8049	-0.34415
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0548294	00020649	W	5.8049	-0.34415
0548294	00020649	W	5.8049	-0.34415
0623715	00033523	W	6.39525	-0.558717
0623715	00033523	W	6.39525	-0.558717
0623715	00033523	W	6.39525	-0.558717
0623715	00033523	W	6.39525	-0.558717
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Subsequently, the dataset underwent a comprehensive data cleaning process. Missing values within repeated sample records were populated using information from other records sharing the same sample identification code, provided that the information was available. Columns containing no observations throughout the dataset or containing a single repeated value that did not contribute meaningful analytical information (e.g.,

Relative Humidity, Calcium, and similar variables) were removed to improve dataset quality and reduce redundancy.

Two samples, **GHM 010** and **GHM 395**, contained extensive missing information, including the sampling location, geographic coordinates, and other essential descriptive variables. Although aflatoxin concentration measurements were available for these samples, they were excluded from subsequent analyses because the absence of spatial location information rendered them unsuitable for spatial analysis and mapping.

Following spatial validation, five samples (**GHG 184, GHM 214, GHM 501, GHG 185, and GHG 191**) were found to plot outside the Ghana national boundary. Since these locations were inconsistent with the documented study area, they were considered spatial outliers and were removed from the final dataset prior to analysis. The resulting cleaned dataset was therefore considered suitable for subsequent spatial mapping, statistical analysis, and geospatial modelling.

In the cleaned Aflatoxin dataset, each sample code had multiple readings (B1, B2, G1, G2, Total Aflatoxin(ppb), etc.). We took the **maximum** aflatoxin value across those replicates for a sample rather than the mean because aflatoxin contamination is notoriously **heterogeneous**. **A mean calculation dilutes these dangerous spikes, hiding localized toxicity that poses severe health risks.**

The depth chosen for the clay and soil pH is the 0–30 cm soil depth is chosen for clay content and soil pH because it represents the primary active root zone and topsoil layer where the toxic fungus *Aspergillus flavus* thrives, interacts with crop roots, and influences maize and groundnut contamination. [more](#) .

Data extraction.

The workflow uses the latitude and longitude coordinates of each sampling point in the Excel file to create precise geographic points in Google Earth Engine. These points are then overlaid on the relevant raster datasets, and the pixel value corresponding to each sample location is extracted. For rainfall, the CHIRPS daily raster data are used to determine the maximum daily rainfall

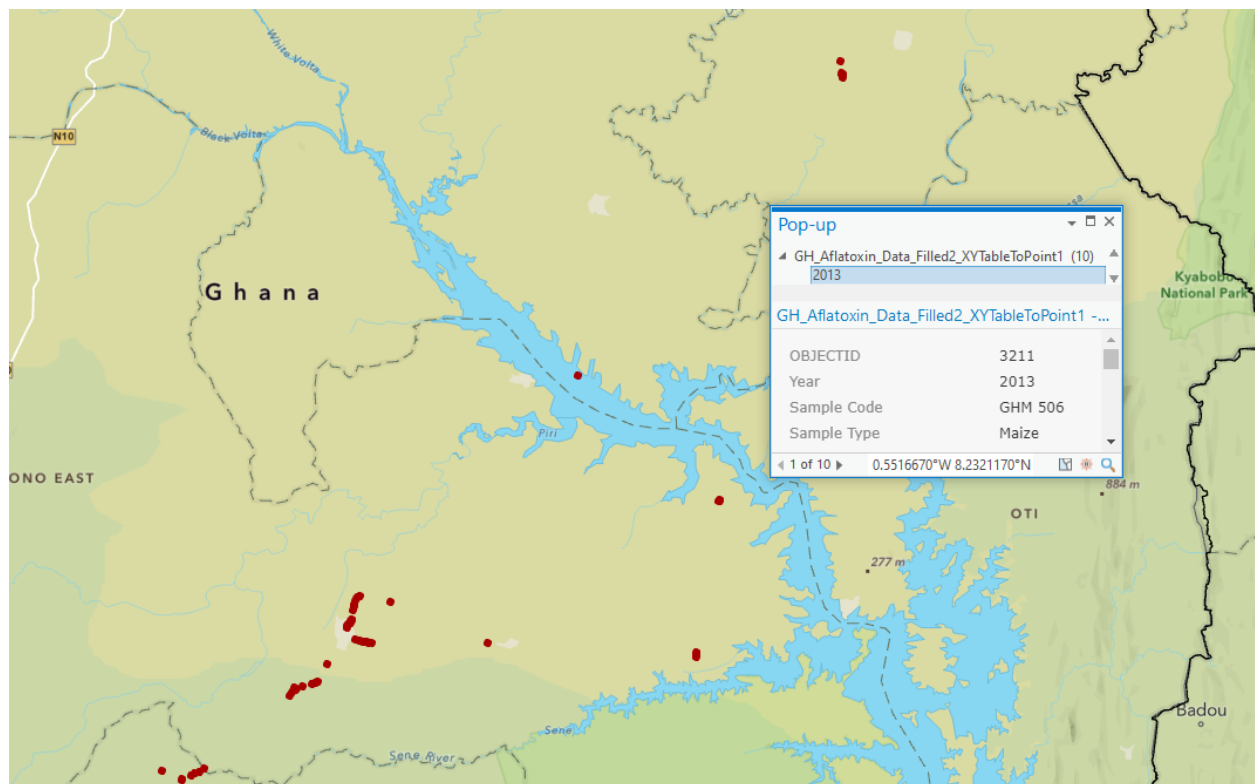
recorded within each month for the sample's specific year and location. Similarly, climate variables such as air temperature, relative humidity, wind speed, surface pressure and soil temperature are extracted from ERA5-Land data, while soil properties such as pH, clay, nitrogen and organic carbon are obtained from the iSDAsoil raster datasets. Therefore, each environmental value in the final dataset is spatially linked to the exact sampling point coordinates provided in your Excel file, with the climate variables organised for each sample's 12 months.

Variable	Unit	Source Dataset	Resolution/Notes
Rainfall (max daily)	mm/day	CHIRPS Daily (UCSB-CHG/CHIRPS/DAILY)	Monthly maximum from daily values, ~5.5 km
Air Temperature (max daily)	°C	ERA5 (temperature_2m)	Converted from Kelvin
Relative Humidity (max daily)	%	ERA5 (derived from temperature_2m & dewpoint_temperature_2m)	Calculated using Magnus formula, clamped 0–100%
Wind Speed (max daily)	m/s	ERA5 (u_component_of_wind_10m, v_component_of_wind_10m)	Resultant speed from u/v components
Surface Pressure (max daily)	hPa	ERA5 (surface_pressure)	Converted from Pa to hPa
Soil Temperature (max daily)	°C	ERA5 (soil_temperature_level_1)	Converted from Kelvin
Soil pH	pH	iSDAsoil (ISDASOIL/Africa/v1/ph)	30 m resolution
Clay Content	g/kg	iSDAsoil (ISDASOIL/Africa/v1/clay_content)	30 m resolution
Nitrogen	g/kg	iSDAsoil (ISDASOIL/Africa/v1/nitrogen_total)	30 m resolution
Organic Carbon	g/kg	iSDAsoil (ISDASOIL/Africa/v1/carbon_organic)	30 m resolution

DUPLICATE COORDINATE ASSESSMENT

The duplicate coordinate assessment identified a few locations where two samples share the same geographic coordinates. These represent different sample types (maize and groundnut) collected from the same farm/location, which is expected because multiple crop types can be sampled at a single location. The only exception is the pair **GHM 037** and **GHM 038**, which share the same coordinates and sample type and should therefore be reviewed for potential duplication.

Additionally, sample **GHM 506** failed to obtain value for several environmental variables, including soil pH, because its location falls within **Lake Volta**, where the corresponding raster datasets contain no valid terrestrial data. Consequently, this sample, together with the five previously identified outlier samples, will be excluded from subsequent analyses to ensure the integrity and reliability of the dataset.



GH AFLATOXIN PROJECT: PYTHON SCRIPT DESCRIPTION

GH_Variable_extraction.py

[GH Variable extraction.py](#)

This script extracts monthly environmental variables for each Ghana aflatoxin sample point using Google Earth Engine. It takes an uploaded Excel file of sample locations (Sample Code, Year, Latitude, Longitude), converts each into a GEE point feature, and pulls monthly maximum rainfall (CHIRPS), air temperature, relative humidity, wind speed, surface pressure, and soil temperature (ERA5), along with static soil properties pH, clay, nitrogen, and organic carbon (iSDAsoil). All extracted values are merged with the original sample metadata into one clean monthly dataset (12 rows per sample), visualized on a Leaflet/folium map, and exported as an Excel workbook.

GH_Variable_Quality_Assessment_and_Similarity_Analysis.py

[GH_Variable_Quality_Assessment_and_Similarity_Analysis.py](#)

This script performs a quality check on the extracted environmental dataset. It loads the dataset and, for each of the 10 environmental variables, reports how many values are missing, unique, or repeated, and lists which sample codes share each value. It also computes descriptive statistics (mean, median, variance) for all variables and identifies samples with identical environmental profiles across all variables combined useful for flagging potential duplicate or suspect extractions. Results are compiled into a multi-sheet Excel workbook (summary, per-variable breakdowns, statistics, and identical profiles).